Performance Evaluation of YOLOv10 for Prohibited Item Detection in X-ray Imagery

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***Abstract*—This paper presents a practical evaluation of the recently proposed YOLOv10, a state-of-the-art, end-to-end object detector. We investigate its practical application for the specialized task of identifying prohibited items in X-ray security scans. This work assesses two model variants, YOLOv10-m (medium) and YOLOv10-l (large), on the Few-Shot Object Detection (FSOD) dataset. A key innovation of YOLOv10 is its NMS-free training strategy; we train both variants for 60 epochs and evaluate them on standard metrics including Precision, Recall, F1-Score, and mean Average Precision (mAP). The YOLOv10-l model achieves the best result, with an mAP@50-95 of 0.555 on the held-out test set (versus 0.541 for YOLOv10-m), providing a strong baseline for the use of end-to-end YOLO models in security contexts.**

***Index Terms*—Object Detection, YOLOv10, NMS-Free, X-ray Security Screening, Deep Learning, FSOD Dataset.**

I. INTRODUCTION

Object detection is a fundamental computer vision task that involves identifying and localizing multiple objects within an image. Its applications are widespread, from autonomous driving to smart healthcare [1]. The "You Only Look Once" (YOLO) series, first proposed by Joseph Redmon, revolutionized the field by reframing detection as a single regression problem, enabling real-time performance [1]. Each subsequent version has improved the critical balance between speed and accuracy.

YOLOv10, introduced by Wang et al. [2], marks a significant leap by creating a truly end-to-end detector. It addresses a key performance bottleneck in previous models: the reliance on Non-Maximum Suppression (NMS) for post-processing. This study provides a practical validation of YOLOv10's architectural innovations by applying it to the challenging domain of X-ray security screening.

II. YOLOv10 ARCHITECTURE AND INNOVATIONS

YOLOv10 achieves its state-of-the-art performance through two primary advancements: a novel training strategy for NMS-free inference and a holistically optimized model architecture [2].

A. NMS-Free Training with Consistent Dual Assignments

A major latency bottleneck in previous YOLO models was the NMS step. YOLOv10 eliminates this by employing a dual-assignment training strategy with two heads:

1) **One-to-Many Head:** Similar to traditional YOLOs, this provides rich, supervisory signals during training, facilitating robust feature learning.
2) **One-to-One Head:** This ensures each ground-truth object corresponds to a single positive prediction. During inference, only this head is used, making the NMS step entirely unnecessary and significantly reducing latency [2].

B. Holistic Efficiency-Accuracy Driven Design

YOLOv10 meticulously re-evaluates and optimizes its components for both speed and accuracy.

- **Lightweight Classification Head:** Recognizing that regression is often more critical for performance, YOLOv10 uses a simpler, lightweight architecture for the classification head, reducing computational overhead.
- **Spatial-Channel Decoupled Downsampling (SCDown):** This replaces standard convolutions with a more efficient method that separates channel transformation and spatial reduction, preserving information while reducing parameters [2].
- **Partial Self-Attention (PSA):** To efficiently model global context, PSA is integrated after the main backbone. It partitions features and applies multi-head self-attention to only one part, significantly reducing the computational cost typically associated with attention mechanisms [2].

III. EXPERIMENTAL SETUP & RESULTS

A. Dataset

The study utilizes the Few-Shot Object Detection (FSOD) dataset, which contains X-ray images of luggage with 20 classes of prohibited or notable items. The dataset is pre-divided into a training set (9,867 images), a validation set (val, 1,233 images, 4,070 object instances), and a held-out test set (newtest, 1,233 images, 3,924 object instances), enabling a robust evaluation of model generalization.

B. Training and Evaluation

Both the YOLOv10-m and YOLOv10-l models were trained for 60 epochs with the Ultralytics implementation on a Kaggle platform using an NVIDIA Tesla T4 GPU, at an input resolution of 640×640 with automatic mixed precision and an automatically selected optimizer. The batch size was 16 for YOLOv10-m and 8 for YOLOv10-l. Performance was evaluated using Precision (P), Recall (R), F1-Score, mAP@50, and mAP@50-95.

C. Performance Analysis

The quantitative results for both models on the validation and newtest sets are presented in Table I. On the held-out newtest set, YOLOv10-l outperforms YOLOv10-m on Precision, F1-Score, mAP@50, and mAP@50-95, while the two models have essentially equal Recall (0.701 versus 0.703). The mAP@50-95 gain from the larger model is modest (0.555 versus 0.541). The close agreement between the validation and newtest scores for both models (for YOLOv10-l, 0.559 on validation versus 0.555 on newtest) indicates that they generalize well and do not suffer from significant overfitting.

Visual analysis of the final model performance on the unseen ‘newtest‘ set provides further insight. The Precision-Recall (PR) curves in Fig. 1 illustrate the trade-off between the metrics, with YOLOv10-l generally maintaining higher precision at various recall levels. The confusion matrices in Fig. 2 identify specific class-level successes and failures. Per-class AP@50 on the newtest set shows the same pattern for both models: strong detection of ‘laptop‘ (0.979 for YOLOv10-m, 0.976 for YOLOv10-l) versus relative difficulty with ‘folding_knife‘ (0.339 and 0.407) and ‘lighter‘ (0.395 and 0.424).

IV. CONCLUSION

This study evaluated the end-to-end YOLOv10 detector on the challenging FSOD X-ray dataset. Both the medium and large variants achieve strong accuracy, with mAP@50 of 0.760 and 0.770 and mAP@50-95 of 0.541 and 0.555, respectively, on the held-out test set. The YOLOv10-l model establishes a solid performance baseline for automated threat detection, although its advantage over YOLOv10-m is modest, and both models struggle with visually ambiguous classes such as folding knives and lighters. Inference latency was not measured in this study, so the speed benefits of the NMS-free design are those reported in [2]; benchmarking latency and addressing the weaker classes are natural directions for future work.

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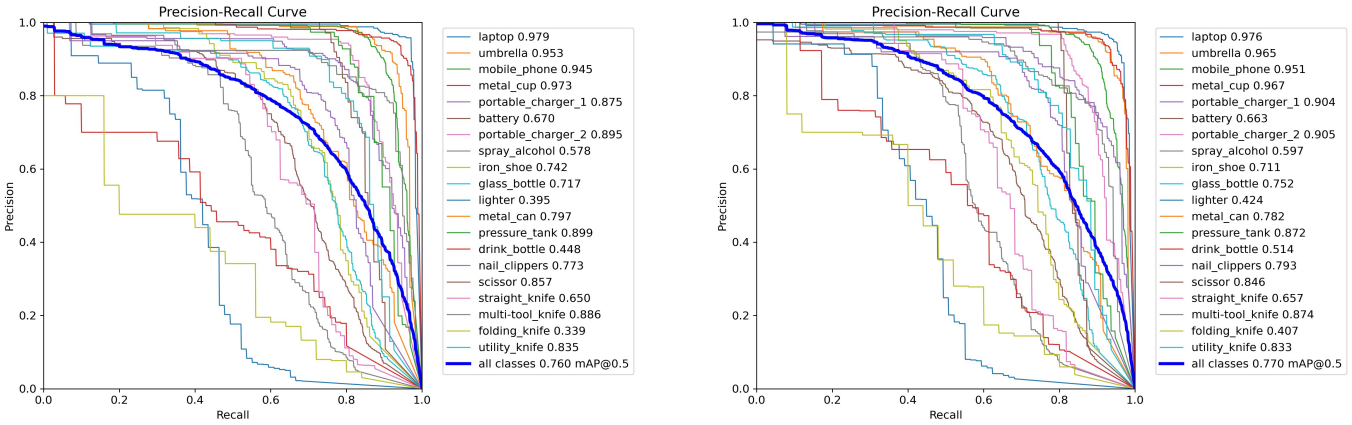
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TABLE I: PERFORMANCE METRICS ON FSOD DATASET

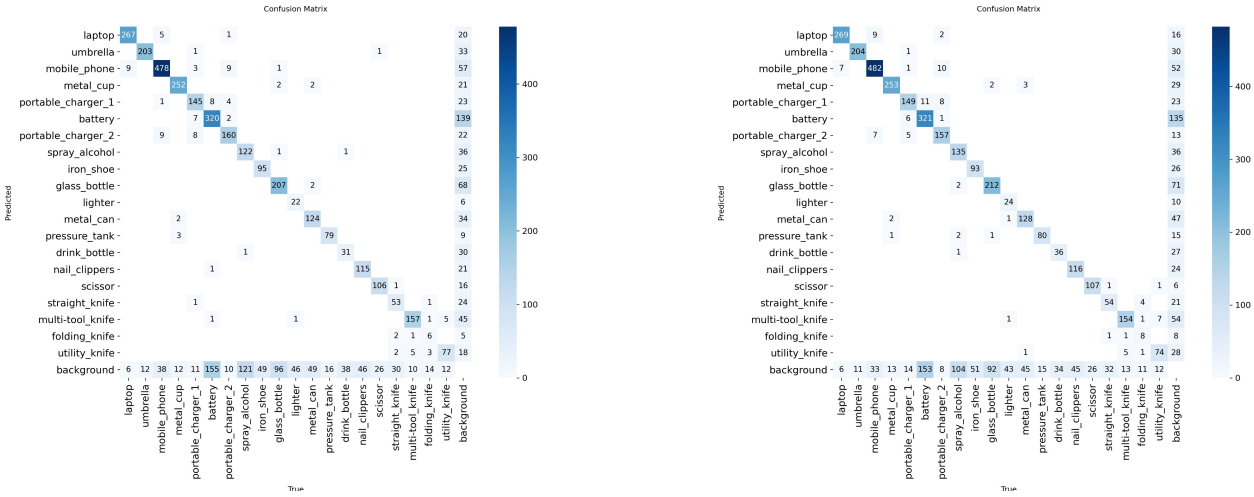
Model	Dataset	Precision	Recall	F1-Score	mAP@50	mAP@.5-.95
YOLOv10-m	Validation	0.817	0.680	0.742	0.762	0.541
	New Test	0.795	0.703	0.746	0.760	0.541
YOLOv10-l	Validation	0.807	0.711	0.756	0.778	0.559
	New Test	0.821	0.701	0.757	0.770	0.555



(a) YOLOv10-m on newtest set

(b) YOLOv10-l on newtest set

Fig. 1: Precision-Recall Curve for (a) YOLOv10-m and (b) YOLOv10-l on the newtest set.



(a) YOLOv10-m on newtest set

(b) YOLOv10-l on newtest set

Fig. 2: Confusion Matrix for (a) YOLOv10-m and (b) YOLOv10-l on the newtest set.